Asymmetric Cryptography

Applied Field Theory 101

Kieran Harvie

Copyright © October 21, 2023. All Rights Reserved.

Contents

1 Introduction **2**
- 1.1 Terminology . . . 2
- 1.2 Trapdoors . . . 2
- 1.3 Field Theory . . . 2

2 Rivest–Shamir–Adleman **3**
- 2.1 Description . . . 3
- 2.2 Applications . . . 3

3 Elliptic Curve Cryptography **4**
- 3.1 Elliptic Curves . . . 4
- 3.2 Elliptic Curve Digital Signature Algorithm . . . 4
- 3.3 Remarks . . . 4

4 Diffie–Hellman Key Exchange **5**

5 Merkle–Hellman Knapsack **6**
- 5.1 Description . . . 7
- 5.2 Remarks . . . 8

A Appendix **9**
- A.1 Algebra Revision . . . 9
  - A.1.1 Lagrange's Theorem . . . 9
  - A.1.2 Cyclic Group . . . 9
  - A.1.3 Bézout's Identity . . . 9
  - A.1.4 Totient Functions . . . 9
- A.2 Subset Sum Problem . . . 10
  - A.2.1 Proof of Correctness . . . 10
  - A.2.2 Proof of Uniqueness . . . 10
- A.3 Montgomery Ladder . . . 10

Introduction

1.1 Terminology

Include public-key encryption and signature table with direction.

Alice, Bob, Mallory.

1.2 Trapdoors

1.3 Field Theory

Rivest–Shamir–Adleman

2.1 Description

2.2 Applications

Elliptic Curve Cryptography

3.1 Elliptic Curves

3.2 Elliptic Curve Digital Signature Algorithm

3.3 Remarks

playstation and Android wallet

Diffie–Hellman Key Exchange

The Diffie-Hellman Key Exchange is a short but sweet demonstration of asymmetric cryptography. The goal is for Alice and Bob to use a public channel to end with a shared private key.

1. Alice and Bob agree on an integer $n > 1$ and a generating element g of the cyclic group C_n in private.
2. Independently and in private, Alice and Bob pick random integers $1 < a < n$ and $1 < b < n$ respectfully.
3. Alice and Bob let g^a and g^b be publicly known.
4. Alice and Bob calculate $(g^b)^a$ and $(g^a)^b$ from the public g^b and g^a respectfully.

Since:

$$(g^b)^a = g^{ba} = g^{ab} = (g^a)^b$$

Alice and Bob agree on g^{ab} while only making n, g, g^a and g^b public. Under the discrete logarithm trapdoor assumption, a third party can not deduce a or b from this information meaning g^{ab} is safe from attackers.

Two important remarks.

Firstly, if g is publicly known then the exchange is susceptible to a man-in-the-middle attack. Mallory is able to pretend to be Alice to Bob and send g^m then Mallory and Bob will share the key g^{mb} with Bob thinking the key was shared with Alice.

Secondly, The most direct implementation uses the $G = (\mathbb{Z}/p\mathbb{Z})^\times$ with prime p and is called the Finite Field Diffie-Hellman. This is because $\mathbb{Z}/p\mathbb{Z}$ is a finite field and $(\mathbb{Z}/p\mathbb{Z})^\times$ is always guaranteed to be a cyclic field.

Merkle–Hellman Knapsack

The knapsack problem is an easy to grasp yet complex to solve optimization problem best summarised through the archetypal example:

> Given a knapsack with with a limited size and a collection of items of fixed size and value determine which selections of items with a combined size less than or equal to the knapsack has the maximum value.

The practical applications are obvious and the problem has been thoroughly studied because of it. Yet the problem remains difficult to compute for arbitrary items, which makes the knapsack problem a candidate for producing a trapdoor function.

For a trapdoor function to work we need a version of the problem which is hard without private information but easy with private information. The Subset Sum Problem (SSP) is a special case of the knapsack problem where the items values are constant and we must find a collection that **equals** the knapsack size, not just be less than. With arbitrary item sizes the SSP is known to be hard, but when the item sizes are superincreasing the problem is trivial.

A sequence $\{w_k\}$ is superincreasing if we have:

$$w_n > \sum_{k=1}^{n-1} w_k$$

In our terms it means there is no combination of items with size less than a given item that have a combined size greater or equal to the first item. This makes solving the superincreasing SSP easy because if we always take the

largest item that would fit in the remaining knapsack size, since the remaining smaller items can't make up the difference.

Hence the plan is to construct some private key which can convert a public SSP to a private superincreasing SSP.

5.1 Description

Key Generation:

Let n be the desired block size.

1. Generate a random superincreasing n-array $\{w_k\}$.
2. Generate a random integer q such that:

$$q > \sum_{k=1}^{n} w_k$$

3. Generate a random integer r that is coprime to q.
4. Calculate a new n-array $\{b_k\}$ such that:

$$b_k = rw_k \mod q$$

5. Calculate r' such that:

$$rr' \mod q = 1$$

The public key is $\{b_k\}$ and the private key is $(\{w_k\}, q, r')$, observe that r can be discarded.

Encryption:

Let the n-bit message be given by the bits m_k. The cipher text is given by:

$$c = \sum_{k=1}^{n} m_k b_k$$

Decryption:

Treat $(r'c \mod q)$ and $\{w_k\}$ as the knapsack size and item sizes for a super-increasing SSP problem, w_k will be in the knapsack if and only if $m_k = 1$.

5.2 Remarks

Appendix

A.1 Algebra Revision

A.1.1 Lagrange's Theorem

A.1.2 Cyclic Group

A.1.3 Bézout's Identity

A.1.4 Totient Functions

A totient of an integer n is a number less than n that is relatively prime to n. Totient are important to cryptography because they are, by definition, the elements the multiplicative group of the ring $\mathbb{Z}/n\mathbb{Z}$, denoted $(\mathbb{Z}/n\mathbb{Z})^\times$.

From this group we define two functions, Euler's totient function:

$$\phi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$$

and the Carmichael function[1]:

$$\lambda(n) = \operatorname{lcm}\left\{ \operatorname{ord}(k) \mid k \in (\mathbb{Z}/n\mathbb{Z})^\times \right\}$$

These functions are often studied in their own right

Chinese Remainder Theorem:

Let n_k be pairwise relatively prime positive integers and let:

$$N = \prod_k n_k$$

[1] I have not idea why Euler gets to own his function but Carmichael does not.

$$\mathbb{Z}/N\mathbb{Z} \cong \prod_k \mathbb{Z}/n_k\mathbb{Z}$$

Euler's Product Formula:

Carmichael's Theorem:

Cyclic Multiplicative group of integers modulo n:

Carmichael's Theorem:

Remark:

A.2 Subset Sum Problem

A.2.1 Proof of Correctness

A.2.2 Proof of Uniqueness

A.3 Montgomery Ladder